WASP-2b

R-0279

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_230803` · created 2026-07-17T14:55:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460159.7963 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.047
Transit depth (Rp/Rs)^2	2.51 +/- 1.5 [%]
Orbital Inclination (inc)	86.33 +/- 2.92
Airmass coefficient 1 (a1)	0.059 +/- 0.037
Airmass coefficient 2 (a2)	2.24 +/- 0.62
Scatter in the residuals of the lightcurve fit is	55.13 %
Best Comparison Star	#1 - [267, 402]
Optimal Aperture	0.87
Optimal Annulus	3.0
Transit Duration (day)	0.076 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.047 vs 0.1326 (deviation 0.36σ)
Duration fit vs published	0.076 d vs 0.0737 d (deviation 0.06σ)
Mid-time O–C	18.4 min
Depth SNR	2.2
Residual scatter	55.13 %

Light curve

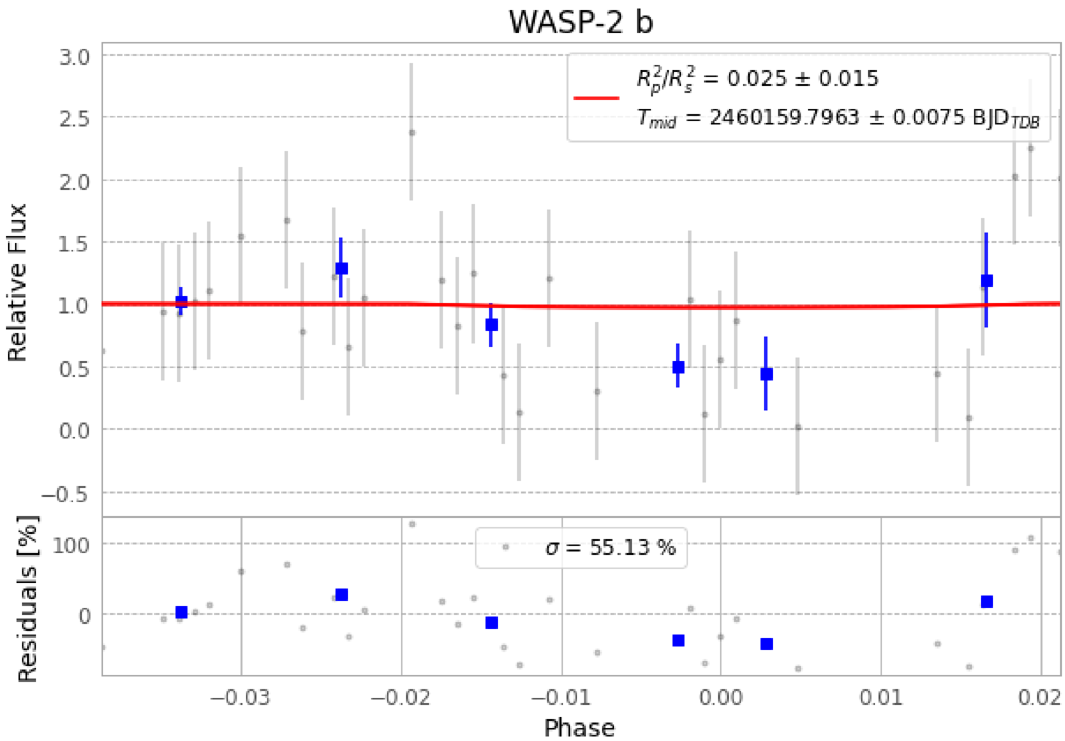

Figure 1 — FinalLightCurve_WASP-2 b_2023-08-02.png (SHA-256 0b3b7013b51fbbba…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 488], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d386202a923f069b…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8d12984`

Data provenance

68 FITS frames (45 MB), `WASP-2230803044815.FITS` → `WASP-2230803080915.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-230803.log (90726582b41e…), FinalLightCurve_WASP-2 b_2023-08-02.png (0b3b7013b51f…), FinalLightCurve_WASP-2 b_2023-08-02.csv (63358fbdfa8d…)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-17 14:55Z.